Nischal Ashok

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RESEARCH INTERESTS

Machine and Deep Learning, Natural Language Processing, Multi-modal Analysis, Explainability and Robustness

EDUCATION BACKGROUND

Indian Institute of Technology (IIT) Patna — **July 2018-May 2022**
Bachelor of Technology (B.Tech), Computer Science and Engineering — ***CGPA: 9.20/10***

PUBLICATIONS

* denotes equal contribution

- Nischal Ashok*, Rina Kumari*, Tirthankar Ghoshal and Asif Ekbal, **"A Multi-task Learning Approach for Fake News Detection: Novelty, Emotion, and Sentiment Lend a Helping Hand"**, International Joint Conference on Neural Networks, **IJCNN-2021**, **[Accepted]**
- Nischal Ashok*, Rina Kumari*, Tirthankar Ghoshal and Asif Ekbal, **"Misinformation detection using multi-task learning with mutual learning for novelty detection and emotion recognition"**, Journal of Information Processing and Management, Elsevier, **IPM-2021**, (IF:4.787) [Paper] [Code]
- Nischal Ashok, Jan Philip Wahle, Terry Ruas, Norman Meuschke, Tirthankar Ghosal and Bela Gipp, **"Empirical Studies on Health Misinformation Detection"**, [under review]
- Rina Kumari, Nischal Ashok, Tirthankar Ghoshal and Asif Ekbal, **"Multi-Modal Dataset for Fake News Detection"**, [under review]
- Nischal Ashok*, Rina Kumari*, Tirthankar Ghoshal and Asif Ekbal, **"What the fake? Probing Misinformation Detection Standing on the Shoulder of Novelty and Emotion"**, Journal of Information Processing and Management, **IPM**, Elsevier (IF:4.787) [under review]
- Nischal Ashok, Venkatesh Babu R, **"Exploring Adversarial Robustness in Deep Neural Networks"**, (Technical Report) Indian Academy of Sciences Research Portal **[Accepted]**

EXPERIENCE

Intern, IBM Research AI — **IBM, Bangalore, India**
Research Managers: • Nitin Gupta • Hima Patel — *May 2021-Present*

- **Project: Data Transformation via Low Code Paradigm**
- Building a reinforcement learning guided deep neural seq-seq approach for automatic program synthesis on semi-structured data.

Undergraduate Researcher, DKE Group — **Universität Wuppertal, Germany**
Advisors: • Dr. Terry Ruas • Prof. Dr. Bela Gipp — *Dec 2020-May 2021*

- **Project: Language Technologies for detecting health fake news**
- Performed and Analyzed Transfer Learning based NLP approaches on health-related fake-news datasets.
- Probed the efficacy of various neural language models with varying training schemes on domain-specific data.

Undergraduate Researcher, AI-NLP-ML Lab — **IIT-Patna, India**
Advisors: • Dr. Asif Ekbal • Prof. Pushpak Bhattacharya — *July 2019-Present*

- **Present Project: Multi-modal Misinformation Detection**
- Probed uni-modal misinformation detection through novelty detection, emotion recognition and sentiment analysis.
- Proposed a novel multi-tasking deep neural network architecture for fake news detection giving State-of-the-Art results on 4 well known datasets.

AI Research Intern, Video Analytics Laboratory — **Indian Institute of Science, India**
Advisor: • Prof. Venkatesh Babu — *May 2020-Aug 2020*

- **Project: Unlabeled Data improves Adversarial Robustness**
- Investigated the role of unlabeled data for improving Adversarial Robustness in Deep Neural Networks using Cifar-10, Cifar-100, and TI-80M as the datasets.
- Probed the role of unlabeled data by proposing new loss functions, sourcing data from unrelated distributions, and generating it from DC-GANs and DE-GANs.
- Showed that unlabeled data from a different distribution helps in improving adversarial robustness.

AI Startup Winter Intern, TildeHat — **Bangalore, India**
Manager: • Dr. Om Deshmukh — *Dec 2019-Jan 2021*

- **Project: Unsupervised Data Retrieval**
- Designed a complete end-to-end module for useful information extraction from resumes.
- Converted pdf and doc files to HTML format and analyzed its DOM structure for information retrieval.

- Implemented unsupervised algorithms and hierarchical classifier approaches for data-retrieval, conducting experiments on thousands of documents.

PROJECTS

IIT-Patna Academic Alexa **IIT Patna**
Guide: Prof. Pushpak Bhattacharya *Jan 2020-May 2020*

- A sentiment-aware intelligent information retrieval based Chatbot for the Academic Portal of IIT-Patna.
- Devised a novel and effective way of converting natural language data into SQL queries using Dependency Tree Parsers, POS tagging, and NER techniques
- Designed a custom chatbot pipeline which includes Multi-Channel CNN based intent classifier, sentiment analyzer, sentence similarity and feedback modules
- Designed a simple portable interface using wx-python GUI programming. [Code] [Blog]

Reddit-Drug Abuse Detection **Data Analytics Lab, IIT Patna**
Guide: Dr. Joydeep Chandra *Jan 2020-May 2020*

- Investigated Drug Abuse on Social Media through Reddit Website by collecting and annotating data from r-opiates sub-reddit.
- Built a multi-head attention deep neural network for classifying comments as motivational or not. Further classification done on altruism, hope, good advice, bad advice, and universality.

Computer Science Academic Projects **IIT-Patna, India**
Guides: • Dr. Jimson Mathew • Dr. Samrat Mondal *May 2020-May 2021*

- **Two Pass Assembler and Emulator** - Built an assembler-emulator system to convert the given assembly language code into a binary object file and then emulate/execute it. [Code]
- **Smart Lock System** - Built a software simulation of a smart lock system using two-step authentication of face recognition (using OpenCV) and pin lock system (implemented in Verilog). [Code]
- **HoneyPot based Authentication System** - Implemented Eugler's HoneyPot based Password Authentication System for providing additional security against traditional password cracking approaches. [Code]

AWARDS and HONORS

- **Conference Grant Scholarships** - ICWSM 2021-AAAI, NAACL-2021
- **Summer Research Fellow** (FASTSF'2020) - Indian Academy of Sciences [India's foremost Science Academy] for pursuing research internship in Artificial Intelligence at IISc, Bangalore.
- **IIT-Patna Merit List** - Among top 10 CPI holder in the IIT-Patna Computer Science class of 2022.
- **Fellowship (KVPY'2016)** - Young Scientist Incentive Plan - Government of India, for pursuing a career in research in pure sciences. Stood among top 0.2% among 200,000 students.
- **Prathiba Ratna Award** - Stood 2nd for Karnataka state in ICSE 10th std Board Exam among 150,000 students.

SKILLS

- **Programming Languages:** • C/C++ • Python • Java • Bash • Verilog • Assembly
- **Deep Learning Tools:** • Tensorflow • Keras • PyTorch • HuggingFace
- **Other tools:** • SQL • Docker • LaTeX • Git/GitHub

SERVICE

- **Volunteer** - NAACL-2021
- **Head of ML and AI** - NJACK - Computer Science Club (IIT-Patna) - Introduced NJACK ML Workshop Series.
- **Departmental Lead** - DSC IIT-P - Associate Facilitator of Explore-ML Workshop IIT-P powered by Google-AI
- **Coordinator** - E-Cell - Introduced Entrepreneur-101 series and organized 4 sessions on entrepreneurship.
- **Mentor** - Institute Student Mentorship Program (IIT-Patna) (2020-Present)
- **Mentor** - NJACK Winter of Code Open Source Programme'2019 - [ML Project Link]
- **Social Worker** - National Service Scheme IIT-Patna
- **Teaching** [Link] and **writing blogs** [Link] on ML and AI